

1. Input:

```
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ nano virtual_address.c
sushmitha@LAPTOP-B907B06L:~$ gcc virtual_address.c -o virtual_address
sushmitha@LAPTOP-B907B06L:~$ ./virtual_address
Value of x = 10
Virtual address of x = 0x7ffffb16fef24
sushmitha@LAPTOP-B907B06L:~$ |
```

2.Input:

```
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address      : %p\n", (void *)main);
    printf("Data address       : %p\n", (void *)&global);
    printf("BSS address        : %p\n", (void *)&bss_variable);
    printf("Heap address       : %p\n", (void *)heap);
    printf("Stack address      : %p\n", (void *)&local);

    free(heap);

    return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ nano memory_segments.c
sushmitha@LAPTOP-B907B06L:~$ gcc memory_segments.c -o memory_segments
sushmitha@LAPTOP-B907B06L:~$ ./memory_segments
Code address : 0x64aae31f41a9
Data address : 0x64aae31f7010
BSS address : 0x64aae31f7018
Heap address : 0x64aae429f010
Stack address : 0x7ffcdc65006c
sushmitha@LAPTOP-B907B06L:~$ |
```

3.Input:

```
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

output:

```
sushmitha@LAPTOP-B907B06L:~$ nano page_table.c
sushmitha@LAPTOP-B907B06L:~$ gcc page_table.c -o page_table
sushmitha@LAPTOP-B907B06L:~$ ./page_table
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
sushmitha@LAPTOP-B907B06L:~$ |
```

4.Input:

```

#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address   = %d\n", virtual_address);
    printf("Page Number       = %d\n", page);
    printf("Offset              = %d\n", offset);
    printf("Frame Number        = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}

```

output:

```

sushmitha@LAPTOP-B907B06L:~$ nano virtual_to_physical.c
sushmitha@LAPTOP-B907B06L:~$ gcc virtual_to_physical.c -o virtual_to_physical
sushmitha@LAPTOP-B907B06L:~$ ./virtual_to_physical
Virtual Address = 250
Page Number    = 2
Offset         = 50
Frame Number   = 1
Physical Address = 150
sushmitha@LAPTOP-B907B06L:~$ |

```

5.Input:

```

#include <stdio.h>
#include <stdlib.h>

int main() {

    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

    free(p);

    return 0;
}

```

output:

```

sushmitha@LAPTOP-B907B06L:~$ nano dynamic_memory.c
sushmitha@LAPTOP-B907B06L:~$ gcc dynamic_memory.c -o dynamic_memory
sushmitha@LAPTOP-B907B06L:~$ ./dynamic_memory
10 20 30 40 50
sushmitha@LAPTOP-B907B06L:~$ |

```

6.Input:

```
#include <stdio.h>

int main() {

    int *p = NULL;

    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}
```

output:

```
sushmitha@LAPTOP-B907B06L:~$ nano null_pointer.c
sushmitha@LAPTOP-B907B06L:~$ gcc null_pointer.c -o null_pointer
sushmitha@LAPTOP-B907B06L:~$ ./null_pointer
Before accessing memory
Segmentation fault      ./null_pointer
sushmitha@LAPTOP-B907B06L:~$ |
```